

3.8 G: Differentiation

	Content
G1	Understand and use the derivative of $f(x)$ as the gradient of the tangent to the graph of $y = f(x)$ at a general point (x, y); the gradient of the tangent as a limit; interpretation as a rate of change; sketching the gradient function for a given curve; second derivatives; differentiation from first principles for small positive integer powers of x Understand and use the second derivative as the rate of change of gradient.
	Content
G2	Differentiate x^n, for rational values of n, and related constant multiples, sums and differences.
	Content
G3	Apply differentiation to find gradients, tangents and normals, maxima and minima and stationary points. Identify where functions are increasing or decreasing.